Muhammad Junaid Ali Asif Raja

Direct PhD Candidate, Computer Science and Information Engineering

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Research Profile

Direct PhD candidate in Computer Science and Information Engineering at National Yunlin University of Science and Technology, advised by Prof. Chuan-Yu Chang. I work in fractional machine learning, an emerging area that brings fractional calculus to optimization, neural architectures, and surrogate modeling for nonlinear dynamical systems, with eight first-author articles among 32 peer-reviewed journal publications since 2024.

Research Interests

- Fractional machine learning: fractional calculus in optimization, neural operators, physics-informed and spiking networks, and surrogates, via memory-aware Caputo, Grünwald–Letnikov, and Riemann–Liouville operators.
- Fractional-order and accelerated optimization algorithms for deep networks.
- Neural surrogate modeling for nonlinear ordinary, delay, and fractional differential systems, including physics-informed networks, neural operators, and autoregressive architectures.
- Embedded and FPGA-aware inference for point-cloud and edge-AI systems.

Education

Direct PhD in Computer Science and Information Engineering — Spring 2024 – Present

National Yunlin University of Science and Technology, Douliu, Taiwan

- Entered the graduate program on the master's track in Spring 2024 and transitioned to the direct PhD track in September 2025.
- Advised by Prof. Chuan-Yu Chang and mentored by Prof. Muhammad Asif Zahoor Raja.
- Current thesis direction focuses on intelligent neural surrogate networks for nonlinear dynamical reconstruction of neuronal systems.

Bachelor's in Electrical Engineering — Sep. 2019 – Jun. 2023

National University of Sciences and Technology (NUST), Islamabad, Pakistan

- GPA: 3.52/4.00. Thesis: "3D Point Clouds on Embedded Platforms."
- Advised by Prof. Faisal Shafait and Prof. Adnan Ul-Hasan; affiliated with the TUKL-DLL Lab, National Centre of Artificial Intelligence.

Research Experience

Scientific Machine Learning Team Lead — Feb. 2025 – Present

Fractional Intelligent Computing Lab, National Yunlin University of Science and Technology, Douliu, Taiwan

- Lead the scientific machine learning subgroup of the Fractional Intelligent Computing Lab, supporting 8+ PhD students across research planning, experiment coordination, and manuscript revision.
- Coordinate collaborative studies under Prof. Muhammad Asif Zahoor Raja on fractional-order optimization and neural modeling of nonlinear dynamical systems.

Graduate Researcher — Feb. 2025 – Present

Intelligent Recognition Industry Service Research Center (IRIS), National Yunlin University of Science and Technology, Douliu, Taiwan

- Conduct IRIS-based research in scientific machine learning and predictive modeling for environmental, biomedical, and cyber-physical systems.

- Collaborate with Chuan-Yu Chang, Prof. Raja, and coauthors on experiment design, validation, and manuscript preparation.

University Research Assistant Jul. 2023 – Present

Future Technology Research Center, National Yunlin University of Science and Technology, Douliu, Taiwan

- Develop implementations of variable fractional-order Adam and generalized fractional stochastic optimization methods.
- Build neural-surrogate workflows for ordinary, delay, and fractional differential systems supporting recommendation and scientific machine learning experiments.

Research Collaborator / Research Member Aug. 2022 – Feb. 2025

Predictive Intelligence and System Modelling, International Islamic University Islamabad, Islamabad, Pakistan

- Studied and implemented probabilistic matrix factorization, neural collaborative filtering, and autoencoder-based recommendation pipelines.
- Contributed to generalized fractional stochastic optimization research and a portable uroflowmetry project with NRTC.

Undergraduate Researcher Mar. 2022 – May 2023

TUKL-NUST Deep Learning Lab, National Centre of Artificial Intelligence, Islamabad, Pakistan

- Supported HLS and FPGA deployment workflows for 3D point-cloud deep-learning models on Xilinx devices.
- Evaluated FINN, BNN, and PointMLP implementation paths for embedded inference and presented the resulting thesis work in seminars, exhibitions, and open-house events.

Mentoring and Leadership

- Lead a scientific machine learning subgroup supporting 8+ PhD students on problem framing, experiment sequencing, result packaging, and manuscript revision, with a research focus on fractional machine learning and intelligent neural surrogates.
- Coordinate multi-author research and manuscript workflows across ongoing projects at YunTech and with earlier research groups.

Selected Publications

Selected journal articles.

1. Adil Sultan, Chuan-Yu Chang, **Muhammad Junaid Ali Asif Raja**, Adiqa Kausar Kiani, Muhammad Shoaib, and Muhammad Asif Zahoor Raja, "Neuro-computational surrogates for aqueous fractional-order nekton-plankton spatiotemporal dynamics under toxicant stress, refuge efficacy, and nutrient flux modulation," *Water Research*, art. 124754, 2026. doi.org/10.1016/J.WATRES.2025.124754
2. Shahzaib Ahmed Hassan, **Muhammad Junaid Ali Asif Raja**, Chuan-Yu Chang, Chi-Min Shu, Muhammad Shoaib, Adiqa Kausar Kiani, Aneela Kausar, and Muhammad Asif Zahoor Raja, "Deep multi-layered autoregressive neuro-structures for predictive modelling of nonlinear chaotic Lorenz-Lü-Chen systems in Rayleigh Bénard convection," *International Journal of Computer Mathematics*, pp. 1–27, 2026. doi.org/10.1080/00207160.2026.2649623
3. Shahzaib Ahmed Hassan, **Muhammad Junaid Ali Asif Raja**, Syed Zoraiz Ali Sherazi, Chuan-Yu Chang, Chi-Min Shu, Adiqa Kausar Kiani, Zeshan Aslam Khan, Muhammad Shoaib, and Muhammad Asif Zahoor Raja, "Design of stochastic backpropagative autoregressive exogenous neuroarchitectures for predictive analysis of fractional-order nonlinear Rabinovich–Fabrikant chaotic attractors," *Nonlinear Dynamics*, vol. 113(25), pp. 34451–34483, 2025. doi.org/10.1007/S11071-025-11827-4
4. **Muhammad Junaid Ali Asif Raja**, Adil Sultan, Chuan-Yu Chang, Chi-Min Shu, Muhammad Shoaib, Adiqa Kausar Kiani, and Muhammad Asif Zahoor Raja, "Design of a fractional-order environmental toxin-plankton

system in aquatic ecosystems: A novel machine predictive expedition with nonlinear autoregressive neuroarchitectures," *Water Research*, vol. 282, art. 123640, 2025. doi.org/10.1016/J.WATRES.2025.123640

5. **Muhammad Junaid Ali Asif Raja**, Adil Sultan, Chuan-Yu Chang, Chi-Min Shu, Adiqa Kausar Kiani, and Muhammad Asif Zahoor Raja, "Bayesian-regularized cascaded neural networks for fractional asymmetric carbon-thermal nutrient-plankton dynamics under global warming and climatic perturbations," *Engineering Applications of Artificial Intelligence*, vol. 151, art. 110739, 2025. doi.org/10.1016/J.ENGAPPAI.2025.110739
6. **Muhammad Junaid Ali Asif Raja**, Zaheer Masood, Ijaz Hussain, Aneela Zameer, and Muhammad Asif Zahoor Raja, "Design of deep learning networks for nonlinear delay differential system for Stuxnet virus spread in an air gapped critical environment," *Applied Soft Computing*, vol. 175, art. 113091, 2025. doi.org/10.1016/J.ASOC.2025.113091
7. **Muhammad Junaid Ali Asif Raja**, Shahzaib Ahmed Hassan, Chuan-Yu Chang, Chi-Min Shu, Adiqa Kausar Kiani, Muhammad Shoaib, and Muhammad Asif Zahoor Raja, "A hybrid neural-computational paradigm for complex firing patterns and excitability transitions in fractional Hindmarsh-Rose neuronal models," *Chaos, Solitons & Fractals*, vol. 193, art. 116149, 2025. doi.org/10.1016/J.CHAOS.2025.116149

Research Projects and Software

zij Open-source PyTorch optimizer library — 100+ optimizers as drop-in replacements for `torch.optim`, drawn from a 740-method canon; `pip install zij`, Apache-2.0. github.com/junaidaliop/zij.

Fractional-order optimization methods Built research implementations of variable fractional-order Adam and generalized fractional stochastic optimization methods used across recommendation, scientific machine learning, and broader large-scale learning studies.

Neural surrogates for differential systems Designed experimental workflows for nonlinear ordinary, delay, and fractional differential systems with applications in neuronal, environmental, biomedical, and cyber-physical modeling.

Recommendation and collaborative filtering Implemented and compared PMF, NCF, user-autoencoder, and item-autoencoder pipelines to study optimization behavior in recommendation settings.

Embedded AI for 3D point clouds Supported HLS and FPGA deployment workflows for point-cloud models on Xilinx devices during the bachelor's thesis project on embedded point-cloud inference.

Honors

Phi Tau Phi Scholastic Honor Society Jun. 2025

National Yunlin University of Science and Technology, Douliu, Taiwan

- Awarded membership in the Phi Tau Phi Scholastic Honor Society of the Republic of China, recognizing the top 2% of graduate master's students for scholastic achievement.

Technical Skills

Programming Python, C++, C, MATLAB, Julia, Bash.

Research areas Fractional-order optimization, accelerated training algorithms, neural surrogate modeling for ordinary / delay / fractional differential systems, physics-informed networks, neural operators, autoregressive architectures, and embedded / FPGA-aware inference.

Tools and workflows PyTorch, Git, Linux, Jupyter, LaTeX, FPGA/HLS workflows.

Full Journal Publication Record

1. Kiran Asma, Muhammad Asif Zahoor Raja, **Muhammad Junaid Ali Asif Raja**, Chi-Min Shu, Muhammad Ali Kiani, and Muhammad Shoaib, "A novel design of time delay fractional nonlinear SIHQR worm transmission model for industrial IoT networks: Machine learning knowledge driven neuroarchitecture analysis," *Chaos, Solitons & Fractals*, vol. 209, art. 118400, 2026. doi.org/10.1016/j.chaos.2026.118400
2. Shehzada Taimur, Muhammad Asif Zahoor Raja, **Muhammad Junaid Ali Asif Raja**, Shahzaib Ahmed Hassan, Sannan Zia Abbasi, Adiqa Kausar Kiani, Muhammad Shoaib, and Chi-Min Shu, "A hybrid intelligent computational framework for diverse firing patterns in a fractional-order Locally Active Memristive Neuron model," *Chaos, Solitons & Fractals*, vol. 208, art. 118209, 2026. doi.org/10.1016/j.chaos.2026.118209
3. Shahzaib Ahmed Hassan, **Muhammad Junaid Ali Asif Raja**, Chuan-Yu Chang, Chi-Min Shu, Muhammad Shoaib, Adiqa Kausar Kiani, Aneela Kausar, and Muhammad Asif Zahoor Raja, "Deep multi-layered autoregressive neuro-structures for predictive modelling of nonlinear chaotic Lorenz-Lü-Chen systems in Rayleigh Bénard convection," *International Journal of Computer Mathematics*, pp. 1–27, 2026. doi.org/10.1080/00207160.2026.2649623
4. Adil Sultan, Chuan-Yu Chang, **Muhammad Junaid Ali Asif Raja**, Adiqa Kausar Kiani, Muhammad Shoaib, and Muhammad Asif Zahoor Raja, "Neuro-computational surrogates for aqueous fractional-order nekton-plankton spatiotemporal dynamics under toxicant stress, refuge efficacy, and nutrient flux modulation," *Water Research*, art. 124754, 2026. doi.org/10.1016/J.WATRES.2025.124754
5. Shahzaib Ahmed Hassan, **Muhammad Junaid Ali Asif Raja**, Syed Zoraiz Ali Sherazi, Chuan-Yu Chang, Chi-Min Shu, Adiqa Kausar Kiani, Zeshan Aslam Khan, Muhammad Shoaib, and Muhammad Asif Zahoor Raja, "Design of stochastic backpropagative autoregressive exogenous neuroarchitectures for predictive analysis of fractional-order nonlinear Rabinovich–Fabrikant chaotic attractors," *Nonlinear Dynamics*, vol. 113(25), pp. 34451–34483, 2025. doi.org/10.1007/S11071-025-11827-4
6. Kiran Asma, Muhammad Asif Zahoor Raja, Chuan-Yu Chang, **Muhammad Junaid Ali Asif Raja**, Chi-Min Shu, and Muhammad Shoaib, "Machine learning knowledge driven investigation for immunity infused fractional industrial virus transmission in SCADA systems," *Journal of Industrial Information Integration*, art. 100940, 2025. doi.org/10.1016/J.JII.2025.100940
7. Adil Sultan, **Muhammad Junaid Ali Asif Raja**, Chuan-Yu Chang, Chi-Min Shu, Adiqa Kausar Kiani, Muhammad Shoaib, and Muhammad Asif Zahoor Raja, "Predictive analysis of plankton population dynamics in marine biosphere: a nonlinear ARX neural network for the carbon-thermal-nutrient-plankton asymmetric multifactor system for global warming," *Environmental Monitoring and Assessment*, vol. 197(12), p. 1367, 2025. doi.org/10.1007/S10661-025-14818-5
8. Hassan Raza, **Muhammad Junaid Ali Asif Raja**, Rikza Mubeen, Zaheer Masood, and Muhammad Asif Zahoor Raja, "Stochastic-Deterministic Modeling of Immune Responses and Tumor Evolution Under Therapeutic Influence: Intelligent Predictive Supervised Exogenous Networks," *IEEE Transactions on Computational Biology and Bioinformatics*, 2025. doi.org/10.1109/TCBBIO.2025.3604337
9. Afshan Fida, Muhammad Asif Zahoor Raja, Chuan-Yu Chang, **Muhammad Junaid Ali Asif Raja**, Zeshan Aslam Khan, and Muhammad Shoaib, "Novel intelligent exogenous neuro-architecture–driven machine learning approach for nonlinear fractional breast cancer risk system," *Communications in Nonlinear Science and Numerical Simulation*, vol. 149, art. 108955, 2025. doi.org/10.1016/J.CNSNS.2025.108955
10. Hassan Raza, **Muhammad Junaid Ali Asif Raja**, Rikza Mubeen, Zaheer Masood, and Muhammad Asif Zahoor Raja, "Supervised autoregressive exogenous networks with fractional Grünwald–Letnikov finite differences: Tumor evolution and immune responses under therapeutic influence fractals model," *Biomedical Signal Processing and Control*, vol. 107, art. 107871, 2025. doi.org/10.1016/J.BSPC.2025.107871
11. Kiran Asma, Muhammad Asif Zahoor Raja, Chuan-Yu Chang, **Muhammad Junaid Ali Asif Raja**, Muhammad Shoaib, and Chi-Min Shu, "A machine learning approach using nonlinear ARX neural networks with Bayesian regularization for epidemic malware dynamics in critical network infrastructures," *International Journal of Information Security*, vol. 24(5), p. 191, 2025. doi.org/10.1007/S10207-025-01104-1
12. **Muhammad Junaid Ali Asif Raja**, Adil Sultan, Chuan-Yu Chang, Chi-Min Shu, Muhammad Shoaib, Adiqa Kausar Kiani, and Muhammad Asif Zahoor Raja, "Design of a fractional-order environmental toxin-plankton system in aquatic ecosystems: A novel machine predictive expedition with nonlinear autoregressive neuroarchitectures," *Water Research*, vol. 282, art. 123640, 2025. doi.org/10.1016/J.WATRES.2025.123640
13. **Muhammad Junaid Ali Asif Raja**, Adil Sultan, Chuan-Yu Chang, Chi-Min Shu, Adiqa Kausar Kiani, and Muhammad Asif Zahoor Raja, "Bayesian-regularized cascaded neural networks for fractional asymmetric carbon-thermal nutrient-plankton dynamics under global warming and climatic perturbations," *Engineering Applications of Artificial Intelligence*, vol. 151, art. 110739, 2025. doi.org/10.1016/J.ENGAPPAI.2025.110739
14. Maryam Pervaiz Khan, **Muhammad Junaid Ali Asif Raja**, Adil Sultan, Chuan-Yu Chang, Muhammad Shoaib, Zeshan Aslam Khan, Adiqa Kausar Kiani, Chi-Min Shu, and Muhammad Asif Zahoor Raja, "Prandtl–Eyring hybrid nanofluidic thermal flow model in solar aircrafts: a novel design of dual-layered nonlinear autoregressive exogenous neural architecture," *Journal of Thermal Analysis and Calorimetry*, vol. 150(13), pp. 10031–10055, 2025. doi.org/10.1007/S10973-025-14396-1
15. **Muhammad Junaid Ali Asif Raja**, Adil Sultan, Chuan-Yu Chang, Chi-Min Shu, Adiqa Kausar Kiani, Muhammad Shoaib, and Muhammad Asif Zahoor Raja, "Prognostication of zooplankton-driven cholera pathoepidemiological Dynamics: Novel Bayesian-regularized deep NARX neuroarchitecture," *Computers in Biology and Medicine*, vol. 192, art. 110197, 2025. doi.org/10.1016/J.COMPBIOMED.2025.110197

16. Roshana Mukhtar, Chuan-Yu Chang, Aqib Mukhtar, **Muhammad Junaid Ali Asif Raja**, Naveed Ishtiaq Chaudhary, Zeshan Aslam Khan, and Muhammad Asif Zahoor Raja, "A novel fractional Parkinson's disease onset model involving α-syn neuronal transportation and aggregation: A treatise on machine predictive networks," *Chaos, Solitons & Fractals*, vol. 194, art. 116269, 2025. `doi.org/10.1016/J.CHAOS.2025.116269`
17. **Muhammad Junaid Ali Asif Raja**, Zaheer Masood, Ijaz Hussain, Aneela Zameer, and Muhammad Asif Zahoor Raja, "Design of deep learning networks for nonlinear delay differential system for Stuxnet virus spread in an air gapped critical environment," *Applied Soft Computing*, vol. 175, art. 113091, 2025. `doi.org/10.1016/J.ASOC.2025.113091`
18. **Muhammad Junaid Ali Asif Raja**, Shahzaib Ahmed Hassan, Chuan-Yu Chang, Chi-Min Shu, Adiqa Kausar Kiani, Muhammad Shoaib, and Muhammad Asif Zahoor Raja, "A hybrid neural-computational paradigm for complex firing patterns and excitability transitions in fractional Hindmarsh-Rose neuronal models," *Chaos, Solitons & Fractals*, vol. 193, art. 116149, 2025. `doi.org/10.1016/J.CHAOS.2025.116149`
19. Adil Sultan, **Muhammad Junaid Ali Asif Raja**, Chuan-Yu Chang, Chi-Min Shu, Adiqa Kausar Kiani, and Muhammad Asif Zahoor Raja, "Predictive modeling of fractional plankton-assisted cholera propagation dynamics using Bayesian regularized deep cascaded exogenous neural networks," *Process Safety and Environmental Protection*, vol. 196, art. 106819, 2025. `doi.org/10.1016/J.PSEP.2025.106819`
20. Hassan Raza, **Muhammad Junaid Ali Asif Raja**, Rikza Mubeen, Zaheer Masood, and Muhammad Asif Zahoor Raja, "Synergistic modeling of hemorrhagic dengue fever: Passive immunity dynamics and time-delay neural network analysis," *Computational Biology and Chemistry*, vol. 115, art. 108365, 2025. `doi.org/10.1016/J.COMPBIOLCHEM.2025.108365`
21. **Muhammad Junaid Ali Asif Raja**, Shahzaib Ahmed Hassan, Chuan-Yu Chang, Chi-Min Shu, Adiqa Kausar Kiani, Muhammad Shoaib, and Muhammad Asif Zahoor Raja, "Design of intelligent Bayesian regularized deep cascaded NARX neurostructure for predictive analysis of FitzHugh-Nagumo bioelectrical model in neuronal cell membrane," *Biomedical Signal Processing and Control*, vol. 101, art. 107192, 2025. `doi.org/10.1016/J.BSPC.2024.107192`
22. **Muhammad Junaid Ali Asif Raja**, Zaheer Masood, Ijaz Hussain, Aneela Zameer, Ammara Mehmood, and Muhammad Asif Zahoor Raja, "Design of nonlinear delay differential system for analyzing vulnerabilities in nanoscale hardware implants: A deep dive into intelligent computing networks," *Nano*, vol. 20(03), art. 2450130, 2025. `doi.org/10.1142/S1793292024501303`
23. Kiran Asma, Muhammad Asif Zahoor Raja, Chuan-Yu Chang, **Muhammad Junaid Ali Asif Raja**, and Muhammad Shoaib, "Machine learning-driven exogenous neural architecture for nonlinear fractional cybersecurity awareness model in mobile malware propagation," *Chaos, Solitons & Fractals*, vol. 192, art. 115948, 2025. `doi.org/10.1016/J.CHAOS.2024.115948`
24. Roshana Mukhtar, Chuan-Yu Chang, Muhammad Asif Zahoor Raja, Naveed Ishtiaq Chaudhary, **Muhammad Junaid Ali Asif Raja**, and Chi-Min Shu, "Design of fractional innate immune response to nonlinear Parkinson's disease model with therapeutic intervention: Intelligent machine predictive exogenous networks," *Chaos, Solitons & Fractals*, vol. 191, art. 115947, 2025. `doi.org/10.1016/J.CHAOS.2024.115947`
25. Zeshan Aslam Khan, Muhammad Waqar, **Muhammad Junaid Ali Asif Raja**, Naveed Ishtiaq Chaudhary, Abeer Tahir Mehmood Anwar Khan, and Muhammad Asif Zahoor Raja, "Generalized fractional optimization-based explainable lightweight CNN model for malaria disease classification," *Computers in Biology and Medicine*, vol. 185, art. 109593, 2025. `doi.org/10.1016/J.COMPBIOMED.2024.109593`
26. Adil Sultan, **Muhammad Junaid Ali Asif Raja**, Chuan-Yu Chang, Chi-Min Shu, Muhammad Shoaib, Adiqa Kausar Kiani, and Muhammad Asif Zahoor Raja, "Intelligent predictive networks for nonlinear oxygen-phytoplankton-zooplankton coupled marine ecosystems under environmental and climatic disruptions," *Process Safety and Environmental Protection*, vol. 193, pp. 733–759, 2025. `doi.org/10.1016/J.PSEP.2024.11.092`
27. **Muhammad Junaid Ali Asif Raja**, Shahzaib Ahmed Hassan, Chuan-Yu Chang, Hassan Raza, Rikza Mubeen, Zaheer Masood, and Muhammad Asif Zahoor Raja, "Novel design of fractional cholesterol dynamics and drug concentrations model with analysis on machine predictive networks," *Computers in Biology and Medicine*, vol. 184, art. 109423, 2025. `doi.org/10.1016/J.COMPBIOMED.2024.109423`
28. Mubeen Sabir, Zeshan Aslam Khan, Muhammad Waqar, Khizer Mehmood, **Muhammad Junaid Ali Asif Raja**, Naveed Ishtiaq Chaudhary, Khalid Mehmood Cheema, Muhammad Asif Zahoor Raja, Muhammad Farhan Khan, and Syed Sohail Ahmed, "Systematic Analysis of Latent Fingerprint Patterns through Fractionally Optimized CNN Model for Interpretable Multi-Output Identification," *Computer Modeling in Engineering & Sciences*, vol. 145(1), p. 807, 2025. `doi.org/10.32604/CMES.2025.068131`
29. Shahzaib Ahmed Hassan, **Muhammad Junaid Ali Asif Raja**, Chuan-Yu Chang, Chi-Min Shu, Muhammad Shoaib, Adiqa Kausar Kiani, and Muhammad Asif Zahoor Raja, "Nonlinear chaotic Lorenz-Lü-Chen fractional order dynamics: A novel machine learning expedition with deep autoregressive exogenous neural networks," *Chaos, Solitons & Fractals*, vol. 189, art. 115620, 2024. `doi.org/10.1016/J.CHAOS.2024.115620`
30. Zeshan Aslam Khan, Muhammad Waqar, Naveed Ishtiaq Chaudhary, **Muhammad Junaid Ali Asif Raja**, Saadia Khan, Farrukh Aslam Khan, Iqra Ishtiaq Chaudhary, and Muhammad Asif Zahoor Raja, "Fractional gradient optimized explainable convolutional neural network for Alzheimer's disease diagnosis," *Heliyon*, vol. 10(20), 2024. `doi.org/10.1016/J.HELIYON.2024.E39037`
31. Nabeela Anwar, **Muhammad Junaid Ali Asif Raja**, Iftikhar Ahmad, and Adiqa Kausar Kiani, "Intelligent heuristic computing paradigm: A novel strategy to singular differential difference equations analysis," *Journal of Innovative Technology*, vol. 6(1), pp. 13–24, 2024. `doi.org/10.70695/JIT.202406_6(1).0002`

32. Nabeela Anwar, **Muhammad Junaid Ali Asif Raja**, Iftikhar Ahmad, and Adiqa Kausar Kiani, "Investigating differential difference equations: an in-depth review," *Journal of Innovative Technology*, vol. 6(1), pp. 25–39, 2024. doi.org/10.70695/JIT.202406_6(1).0003